1 FIG-004 — Current A–G multi-seed audit

Group: MD-1NS **Evidence chain:** manifested **Status:** publication candidate

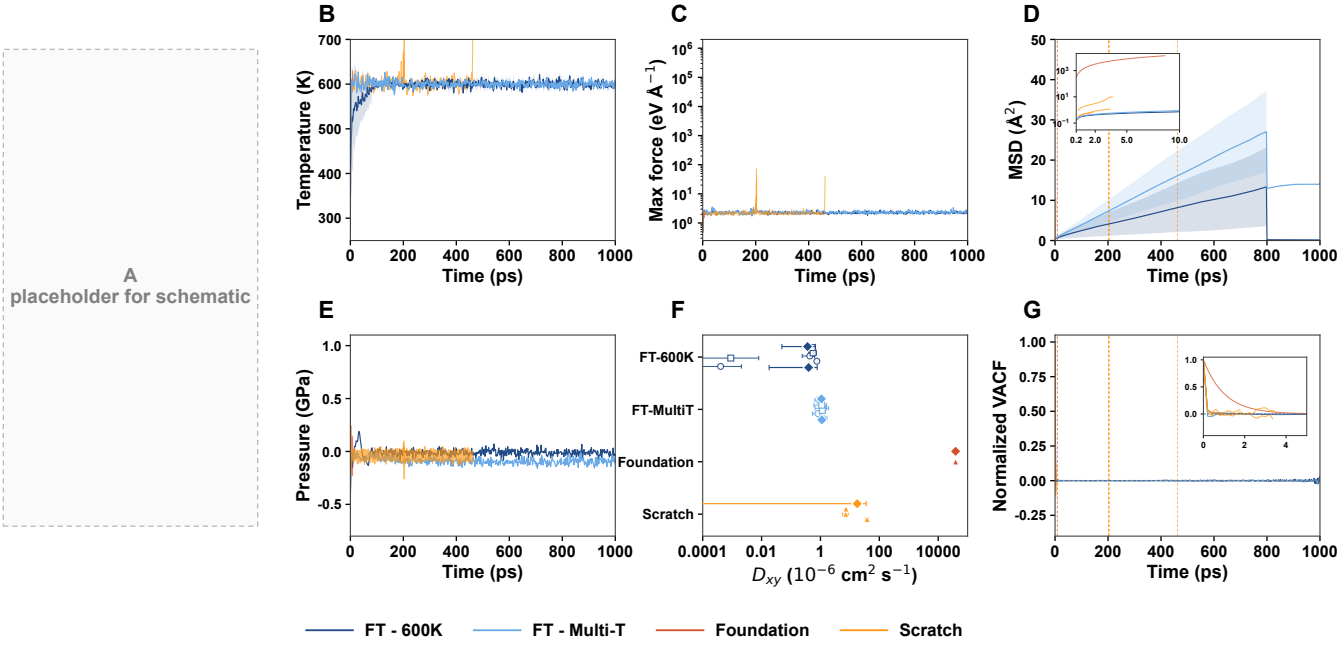


Figure 1: **Diffusion-focused benchmarking of four MACE model families in 1 ns molecular dynamics.** (A) Schematic placeholder retained for the approved page structure. (B) Temperature. (C) Maximum force on a logarithmic axis. (D) Cr mean-squared displacement, with a 0–10 ps log inset to retain the early failure branches alongside the main 0–1000 ps view. (E) Pressure derived only from real periodic-cell stress metadata. (F) Per-training-seed D_{xy} estimates for the 200–500 ps and 500–900 ps windows, together with short pre-failure fits where an MSD source exists; point intervals are shown for each fit and diamonds are family summaries. (G) Long-time normalized velocity autocorrelation function with a 0–5 ps inset. Solid family lines are means across provenance-matched stable training seeds; translucent bands are the corresponding seed standard deviations. Failure-mode trajectories are shown only before the first scientific instability and are excluded from stable-family means. Short pre-failure D_{xy} values are not 1 ns transport coefficients. The shared MD velocity seed is recorded separately and is not an independent replica matrix. Heat-flux autocorrelation and thermal conductivity are not claimed because no matching per-frame heat-flux observable is present.

Results text draft. The current FIG-004 release uses the approved A–G page structure. Panels B–E and G show stable-family means with translucent standard-deviation bands when multiple provenance-matched training seeds cover the same observable; singleton pressure traces remain visible without fabricated uncertainty. Panel F reports each stable FT training seed and its family summary for the 200–500 ps and 500–900 ps windows, together with short pre-failure MSD fits for failure branches that have position data. The new from-scratch seeds 123, 234, and 345 fail the smoke scientific gate near 202 ps; their short pre-failure D_{xy} values are retained separately and are not interpreted as 1 ns transport coefficients. **Discussion and boundary.** The figure supports dynamic-stability and diffusion discussion under an evidence-bounded interpretation. Failure-mode trajectories are retained only before their first scientific threshold event: Foundation-OMAT at 8.8 ps, the legacy from-scratch branch at 462.2 ps, and the new all-data branches at approximately 202.0–203.6 ps. Short pre-failure MSD fits are shown where position data exist; they are excluded from stable-family means and are not 1 ns transport summaries. Pressure is drawn only where real stress metadata is available. HFACF and thermal conductivity remain unavailable because no matching per-frame heat-flux series was recorded. **Article placement.** Proposed article location: Results: dynamic stability and diffusion; SI transport candidate. **Reviewer linkage:** overall status PENDING.

- **REV-C9 (PENDING)**: Keep the transport comparison bounded to the matched, provenance-labelled rows. *Completed:* FT-600K and FT-MultiT now show three training-seed rows and family uncertainty in the current FIG-004; failure branches with archived positions now contribute short pre-failure MSD fits only. *Still required:*

resolve the historical model-hash gaps and preserve the explicit absence of a heat-flux observable before making any κ claim.

Evidence pointers.

- Current context and gates: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_all_data_v1/md_1ns_monitor_context.md
- Additive release manifest: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_merged_20261016/merge_manifest.json
- Plot/processing script: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_merged_20261016/render_fig004.py
- Audit script: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_merged_20261016/audit_from_scratch.py
- Model registry and failure ledger: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_merged_20261016/MODEL_REGISTRY.tsv; /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_merged_20261016/FAILURE_EXCLUSION.tsv
- Pre-failure observables and short-lag D_{xy} fits: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_merged_20261016/PREF_FAILURE_METRICS.tsv
- Claim-to-evidence matrix: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_merged_20261016/CLAIM_EVIDENCE_MATRIX.tsv

Machine-collected evidence candidates for the current FIG-004 contract.

- **Current context:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_all_data_v1/md_1ns_monitor_context.md
- **Release manifest:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_merged_20261016/merge_manifest.json
- **Render manifest:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_merged_20261016/FIG004_RENDER_MANIFEST.json
- **Evidence matrix:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_merged_20261016/CLAIM_EVIDENCE_MATRIX.tsv
- **Failure/exclusion ledger:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/mace_1ns_merged_20261016/FAILURE_EXCLUSION.tsv

Audit gap / next action: Seed-123 failure trajectory copy and historical model hashes remain provenance gaps; retain the explicit no-HFACF boundary.